RUNSHI TANG

rtang56@wisc.edu ⋄ Madison, WI ⋄ October 21, 2025

EDUCATION

Bachelor of Science 2016-2020
School of Mathematics, Shandong University
-Academic Excellence Award in 2016-2017, 2017-2018
(Exchanged to UW-Madison for further education in advance since 2019)

Master in Statistics - Data Science Track 2019-2021
Department of Statistics, University of Wisconsin-Madison
-Academic Excellence Award in 2019-2020

Ph.D. in Statistics 2021-Present
Department of Statistics, University of Wisconsin-Madison; Advised by Profs. Anru Zhang and Richard Chappell

PAPERS

(†: equal contribution)

1. **Tang, R.**, Yuan, M., & Zhang, A. R. (2025). Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model. *Journal of the Royal Statistical Society, Series B*, 87, 232-255.
2. **Tang, R.**, Kolda, T., & Zhang, A. R. (2025+). Tensor Decomposition with Unaligned Observations. *SIAM Journal on Matrix Analysis and Applications*, to appear.
3. Towe, S. L., **Tang, R.**, Gibson, M. J., Zhang, A. R., & Meade, C. S. (2023). Longitudinal changes in neurocognitive performance related to drug use intensity in a sample of persons with and without HIV who use illicit stimulants. *Drug and alcohol dependence*, 251, 110923.
4. Zhang, A. R.†, Bell, R.†, An, C.†, **Tang R.**†, Shana Hall, S., Chan, C., Al-Khalil, K., & Meade, C. (2023). Cocaine Use Prediction with Tensor-based Machine Learning on Multimodal MRI Connectome Data. *Neural Computation*, 36, 107-127.
5. Ko, A., Berger, L., Brown, D., Pac, J., & **Tang, R.** (2025). Unemployment Insurance Generosity and Child Protective Services Involvement *Child Abuse & Neglect*, 167, 107583.
6. **Tang, R.** Chhor, J., Klopp, O., & Zhang, A. R. (2025+). Revisit CP Tensor Decomposition: Statistical Optimality and Fast Convergence. *The Annals of Statistics*, submitted.
7. **Tang, R.**† Zheng, R.†, Zhang, A. R., & Priebe C. (2025+). A functional tensor model for dynamic multilayer networks with common invariant subspaces and the RKHS estimation. *Journal of Computational and Graphical Statistics*, submitted.
8. McCann, J. R., Yang, C., Bihlmeyer, N., **Tang, R.** , et al. (2025+) Branched chain amino acid metabolism and microbiome in adolescents with obesity during weight loss therapy *Submitted.*
9. Pac, J., Berger, L., Brown, D., & **Tang, R.** (2025+). Does Public Program Participation Impact Child Protective Services Involvement? Quasi-Experimental Evidence from the Staggered eWIC EBT Rollout. *Working paper*
10. Pac, J., Berger, L., Durrance, C., Nobles, J., & **Tang, R.** (2025+) Foster Care in the First Year of Life: Effects on Health and Safety. *Working paper*
11. **Tang, R.** Chhor, J., Klopp, O., Tsybakov, A., & Zhang, A. R. (2025+). Density Estimation for the Multiview Model. *Working paper*
12. **Tang, R.** Han, Y., & Zhang, A. R. (2025+). High-order moments and cumulants estimation. *Working paper*

CONFERENCE/WORKSHOP ABSTRACTS

1. McCann, J., Bihlmeyer, N., Yang, C., Truong, T., **Tang, R.** et al. (2024). Unique Age and Sex Dependent Serum Metabolite Signature in Teens With Obesity. *Abstracts from the 42nd annual meeting of the Obesity Society at Obesity Week November 3-6, 2024*

2. Pac, J., Berger, L., Durrance, C., Nobles, J., & **Tang, R.** (2025) Foster care in the First Year of Life: Effects on Health and Safety. *2024 Association for Public Policy Analysis and Management* and *2025 Atlanta Workshop on Public Policy and Child Well-being*
3. Pac, J., Pilarz, A., Slopen, M., Brook, S., & **Tang, R.** (2026) Collective impacts: Paid Family and Medical Leave interacts with state policies to predict maternal employment *2026 Work and Family Researchers Network Conference*

AWARDS

1. Honorable Mention Award in the 2024 Student Paper Competition of the ASA Statistical Learning and Data Science Section, "Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model."
2. Student and Early Career Travel Fund by ASA
3. SRGC Conference Presentation Award by the University of Wisconsin-Madison
4. Travel Scholarship by Sandra Rosenbaum School of Social Work & Institute for Research on Poverty, UW-Madison
5. Hannan Graduate Student Travel Award by IMS

TALKS AND PRESENTATIONS

1. The 18th annual CFAR Fall Scientific Retreat, "Longitudinal changes in neurocognitive performance related to drug use intensity in a sample of persons with and without HIV who use illicit stimulants. "
2. The 37th New England Statistics Symposium, "Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model."
3. 2024 Joint Statistical Meetings, "Mode-wise Principal Subspace Pursuit and Matrix Spiked Covariance Model."
4. 2025 Joint Statistical Meetings, "Revisit CP Tensor Decomposition: Statistical Optimality and Fast Convergence."

EXPERIENCES

Project Assistant Sep 2022 - Now
Institute for Research on Poverty, UW-Madison *Madison, WI*

- Supervised by Prof. Jessica Pac and Prof. Lawrence Berger (School of Social Science & Institute for Research on Poverty, UW-Madison)
- Study the relationship between child well-being and various policies
- Part of our results were presented by Prof. Lawrence Berger in the 2023 International Society for the Prevention of Child Abuse & Neglect Congress

Academia Internship Summer, 2022
School of Medicine, Duke University *Durham, NC*

- Supervised by Prof. Anru Zhang (Department of Biostatistics & Bioinformatics) and Prof. Christina Meade (Department of Psychiatry and Behavioral Sciences)
- Participated in two projects: longitudinal study on HIV contributes to neurocognitive impairments in persons who use stimulants and applying the novel tensor-based method on MRI data to predict cocaine use
- Presented a poster at the 18th annual CFAR Fall Scientific Retreat

Teaching Assistant Sep 2021 - May 2022
Department of Statistics, UW-Madison *Madison, WI*

- Fall 2021: STAT 324, Introductory Applied Statistics for Engineers
- Spring 2022: STAT 324, Introductory Applied Statistics for Engineers

Project Assistant Sep 2020 - Aug. 2021
Department of Statistics, UW-Madison *Madison, WI*

- Supervised by Prof. Awad Hanna (Department of Civil & Environmental Engineering, UW-Madison) and Prof. Wei-Yin Loh (Department of Statistics, UW-Madison)
- Using statistics and programming, Process and analyze data for a research project on construction material transportation of the Department of Civil & Engineering.

Actuarial Internship Summer, 2018
Taishan Property Insurance Co., Ltd. *Jinan, Shandong Province, China*

- Work for Actuarial Dept.; Learn structure of insurance company and knowledge of Property Insurance and making Actuarial Report; Help make observation report on opponents.

RESEARCH INTEREST

- High dimensional statistics, spiked covariance model, tensor decomposition, matrix analysis, optimization ...
- Applying theories into different practical fields, including social science and clinical science.